Introduction to Statistical Methods — Quiz Solutions

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Q1. Two cards are drawn from a deck without replacement. What is the probability that both are face cards, given that the first is a heart?

a. 1/13 b. 5/26 c. 3/26 d. None

Answer: d. None

Explanation:

Given the first card is a heart, there are 13 hearts, of which 3 are face cards (J, Q, K of hearts). P(1st card is a face card | heart) = 3/13. After removing that card, 51 cards remain with 11 face cards left. P(2nd is also a face card) = 11/51. Multiplying: (3/13) × (11/51) = 11/221. Since 11/221 does not match any given option, the answer is 'None'.

Q2. A real estate company has two contractors (A and B). P(assigned to A) = 0.7, P(assigned to B) = 0.3. P(A fails) = 0.02, P(B fails) = 0.04. A project is a loss when the assigned contractor fails. What is the probability that Contractor A is involved in the project failure?

a. 7/12 b. 14/260 c. 7/13 d. 6/13

Answer: c. 7/13

Explanation:

This is Bayes' theorem: find P(A | failure). P(failure) = P(A)×P(fail|A) + P(B)×P(fail|B) = (0.7×0.02) + (0.3×0.04) = 0.014 + 0.012 = 0.026. P(A | failure) = P(A)×P(fail|A) / P(failure) = 0.014 / 0.026 = 14/26 = 7/13.

Q3. Which of the following will increase the value of mean by 1?

a. Adding 1 to the highest value only b. Adding 1 to all the values c. Adding 1 to the lowest value only d. Adding 1 to both highest and lowest values

Answer: b. Adding 1 to all the values

Explanation:

Mean = (sum of values)/n. Adding 1 to only one or two values increases the sum by a small fixed amount that gets diluted by n, not raising the mean by exactly 1 (unless n=1 or 2, not general). Adding 1 to every value increases the total sum by n, so the new mean = (sum + n)/n = old mean + 1. This is the only option guaranteed to raise the mean by exactly 1 for any dataset size.

Q4. A random variable X has the distribution: P(X=-3)=k, P(X=-2)=0.1, P(X=-1)=k, P(X=0)=0.2, P(X=1)=2k, P(X=2)=0.4, P(X=3)=2k. Find the value of k.

a. 0.5 b. 0.05 c. 0.07 d. None

Answer: b. 0.05

Explanation:

All probabilities must sum to 1: k + 0.1 + k + 0.2 + 2k + 0.4 + 2k = 1. Combine: 6k + 0.7 = 1, so 6k = 0.3, giving k = 0.05.

Q5. A card is drawn at random from a standard 52-card deck. Given that it is a face card, what is the probability that it is a king?

a. 1/3 b. 1/13 c. 4/13 d. 1/4

Answer: a. 1/3

Explanation:

There are 12 face cards total (J, Q, K in each of 4 suits), and 4 of them are kings. P(king | face card) = 4/12 = 1/3.

Q6. If X is a random variable such that P(X=1) = 0.3 and P(X=4) = 0.7, then E[X] will be?

a. 2.5 b. 3.5 c. 4 d. 3.1

Answer: d. 3.1

Explanation:

$E[X] = \Sigma\, x \cdot P(x) = (1 \times 0.3) + (4 \times 0.7) = 0.3 + 2.8 = 3.1$.

Q7. An online store has 8 white, 5 blue, and 7 black shirts (20 total). Customer 1 buys a blue shirt. Customer 2 buys a shirt immediately after, but the website didn't update the stock in time. What is the probability that Customer 2 also buys a blue shirt?

a. 5/20 b. 3/20 c. 2/10 d. 4/19

Answer: a. 5/20

Explanation:

Because the website failed to update, Customer 2 is choosing from the stock as it was originally displayed: 20 total shirts with 5 blue shirts still shown as available. So P(blue) = 5/20 (this differs from the 'true' updated probability of 4/19, which would apply only if the site had updated correctly).

Q8. Two dice are thrown simultaneously. What is the probability of obtaining a total score less than 5?

a. 1/6 b. 5/36 c. 5/6 d. 9/36

Answer: a. 1/6

Explanation:

Total outcomes = 36. Sums less than 5 are 2, 3, and 4. Sum=2: (1,1) $\rightarrow$ 1 way. Sum=3: (1,2),(2,1) $\rightarrow$ 2 ways. Sum=4: (1,3),(3,1),(2,2) $\rightarrow$ 3 ways. Total favorable = 1+2+3 = 6. Probability = 6/36 = 1/6.

Q9. An experiment yields three mutually exclusive events A, B, C such that P(A) = 2P(B) = 3P(C). Find P(A■).

a. 1/3 b. 6/11 c. 5/11 d. 4/5

Answer: c. 5/11

Explanation:

Let P(C) = x. Then P(B) = (3/2)x is wrong direction — instead set common value: P(A)=2P(B)=3P(C)=k, so P(A)=k, P(B)=k/2, P(C)=k/3. Since A, B, C are mutually exclusive and exhaustive (only these three events), their probabilities sum to 1: k + k/2 + k/3 = 1 → (6k+3k+2k)/6 = 1 → 11k/6 = 1 → k = 6/11. So P(A) = 6/11, and P(A■) = 1 - 6/11 = 5/11.

Q10. What is the first quartile (Q1) of this data set? Data: 5, 7, 9, 11, 13, 15, 17

a. 11 b. 5 c. 7 d. 9

Answer: c. 7

Explanation:

The data (n=7) is already sorted. Q1 is the median of the lower half (excluding the overall median if n is odd). Lower half = {5, 7, 9}. The median of this lower half is 7. So Q1 = 7.